



Graph neural network-enhanced recursive LSTM for bidirectional seismic response prediction of steel framed structures

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ABSTRACT

This study proposes a novel graph neural network (GNN)-enhanced recursive long short-term memory (LSTM) model, “GNNrLSTM”, for the efficient prediction of bidirectional seismic responses in three-dimensional steel framed structures. The core innovation lies in synergistically integrating a GNN module, which encodes complex structural topologies into generalizable representations, with a recursive LSTM that captures temporal dynamics. This approach overcomes a critical limitation of prior models by enabling a single, unified framework to generalize across diverse structural configurations within the predefined design domain, rather than being confined to a specific structure. A two-stage corrective training strategy is introduced to mitigate error accumulation inherent in recursive predictions. Validated on a dataset of 4000 nonlinear analyses across 128 structures, the GNNrLSTM achieves high prediction performance ($R^2 > 0.95$ for displacement and acceleration) while drastically reducing computational time compared to conventional finite element simulations. By enabling rapid prediction of full response time histories, this approach offers a practical tool for computationally demanding tasks like seismic resilience assessment, highlighting the potential of integrating graph-based spatial encoding with recurrent temporal modeling.

1. Introduction

In recent decades, seismic design has increasingly shifted from focusing solely on collapse prevention and life safety to embracing a more comprehensive concept of “seismic resilience”, which encompasses social, environmental, and economic factors. This trend is highlighted by recent major earthquakes, notably the 2011 Christchurch earthquake in New Zealand. In its aftermath, many code-compliant buildings were demolished due to high repair costs, while a local hospital designed with seismic resilience principles quickly resumed operations. As global attention on seismic resilience grew, many countries developed standards or guidelines to implement this new paradigm, such as China’s GB/T 38591–2020 [1] and the US FEMA P-58 [2]. However, several challenges still impede the wider adoption of seismic resilience assessment. A key obstacle is the high computational cost of nonlinear time history analyses (NTHA), often required to assess the resilience of individual structures or communities. These assessments typically involve numerous NTHA, making widespread implementation difficult.

As a result, developing efficient surrogate models for parts or the entirety of the resilience assessment process has become a critical focus. Machine learning (ML) techniques, renowned for their exceptional data-fitting capabilities, have emerged as a preferred solution for addressing this challenge. Some studies bypassed the “structural analysis” step, directly mapping design information to the damage states or even final resilience metrics [3–12]. While this approach is undoubtedly straightforward, it is also highly challenging and prone to accuracy loss due to error accumulation [13]. Therefore, the focus of existing studies is more on developing surrogate models for the most time-consuming part, NTHA. For example, Guan et al. [14] utilized four ML algorithms, i.e., multivariate linear regression, kernel ridge regression, XGBoost, random forest, and artificial neural network (ANN), to predict peak drift demands of planar steel moment frame buildings, while Nguyen et al. [15] enabled similar predictions by using ANN and extreme gradient boosting. Hu et al. [16–18] and Zhang et al. [19,20] realized the prediction of peak drift, residual drift, and acceleration responses of self-centering steel structures based on an ANN or a deep learning neural network.

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These studies mainly focused on predicting a single response (e.g., peak or residual value) rather than a complete response time series.

While current seismic resilience assessment methodologies primarily rely on peak and residual demands to evaluate the corresponding damage state, it would be desirable to predict the entire seismic response history with comparable computational efficiency, which provides a more comprehensive depiction of the structure's behavior. Achieving this level of prediction typically relies on the capabilities of deep learning, as traditional ML methods often fall short in handling the high-dimensional, nonlinear, and sequential characteristics inherent in ground motion data and seismic response data. Zhang et al. [21] introduced the law of dynamics as physics constraints in a convolutional neural network (CNN) to model seismic response. Liao et al. [22] combined the attention mechanism with LSTM to develop an "AttLSTM" network for bridge seismic response modeling. Wang et al. [23] proposed a novel "deep structural nonlinear analysis" framework for predicting various mechanical responses, and validated it with steel plate shear wall structures. Li et al. [24] proposed a novel structural temporal fusion network to improve the generalization of the model to different structural geometric properties. Ning et al. [25] investigated the use of three different deep learning models to predict nonlinear seismic response on three specific structures. Zhang et al. [26] proposed the "Patch-fusion" deep learning framework, establishing a new state-of-the-art performance by effectively fusing static and dynamic features using novel patching and attention mechanisms. These studies demonstrate the strong potential of deep learning for seismic response history prediction through different network architectures and output strategies. However, for many direct sequence-mapping approaches that predict response histories from ground-motion records, long and high-dimensional response sequences may require fixed-length input/output formatting, sufficient receptive fields, or relatively complex network architectures. This, combined with limited data, may lead to overfitting and hinder generalization, particularly for high-frequency components or predictions with variable lengths [27]. The recursive LSTM, inspired by the stepwise integration method in structural dynamics, proposed by Xu et al. [27] is a feasible solution to improve this problem. It should be noted that "recursive" here refers to the prediction strategy rather than the inherent recurrent architecture of the LSTM cell itself. Specifically, the LSTM is used as a local predictor, and previously predicted responses are recursively fed back together with recent excitation and response information to generate subsequent responses. The proposed recursive LSTM employs a simple, flexible architecture to predict nonlinear structural seismic responses of arbitrary lengths and sampling rates, with LSTM significantly enhancing computational efficiency by focusing solely on target node responses and discarding unnecessary structural parameters. Furthermore, the majority of existing studies on seismic time history prediction for steel frames, regardless of whether they employ the direct sequence-mapping or recursive methods, are typically based on single-structure case studies. Consequently, the resulting models often learn only the unique dynamic characteristics of one specific building typology, which means that the model needs to be retrained for another structure. The few studies that do consider geometric generalization have predominantly focused on 2D planar structures. This simplification may be pertinent for design practices in regions like the United States, where lateral-force-resisting frames are often designed separately from the gravity-framing system. However, this 2D simplification is often inappropriate for design philosophies prevalent in countries like China, where the entire 3D structure, encompassing both perimeter and interior frames, is designed to act as a monolithic system in resisting combined lateral and gravity loads.

Therefore, this study aims to enhance the recursive LSTM network by integrating a graph neural network (GNN) module, where the GNN is utilized to process graphs that represent the structural characteristics. GNNs are a class of neural networks specifically designed to process and learn from graph-structured data, which resides in non-Euclidean space and is challenging for other neural networks to handle. By iteratively aggregating messages from neighboring nodes, GNNs can

effectively capture complex dependencies and structural patterns within the graph, making them particularly well-suited for tasks involving relational or irregular data. Currently, GNNs have shown their advantages in various tasks in the field of structural engineering, such as response prediction and running safety assessment of train-bridge coupled systems [28–30], structural component design [31–36], progressive collapse assessment [37], imperfect single layer gridshells' nonlinear buckling capacity prediction [38], elastic structural analysis [39], static structural analysis [40], and seismic response prediction [41–43]. Among these studies, Kuo et al. [42] proposed a GNN-LSTM-based fusion model to enhance the generalization capability of LSTM-based predictors across different structural configurations. Therefore, the general idea of combining GNN-based structural encoding with LSTM-type temporal modeling has already been explored. However, the coupling of GNN-derived structural representations with a recursive LSTM rollout predictor remains insufficiently investigated. Meanwhile, although the recursive LSTM has shown advantages in avoiding direct long-sequence mapping through stepwise prediction, it has previously been mainly demonstrated for single-structure response prediction and lacks explicit structural topology encoding. To bridge these two lines of work and address the bidirectional response-history prediction requirement in GB/T 38591–2020 [1], this study integrates GNN-based structural graph encoding with a recursive LSTM rollout framework for three-dimensional steel frames.

In light of the above, this paper proposes a novel GNN-enhanced recursive LSTM (GNNrLSTM) network designed for the bidirectional seismic response prediction of three-dimensional steel frames. The structure of the paper is organized as follows: Section 2 outlines the archetype and conceptual framework of the proposed GNNrLSTM network. Section 3 provides a comprehensive description of the structural database generation process (e.g., steel framed structures generation, ground motion selection and nonlinear time history analysis), and the experimental setup. The model performance is evaluated in Section 4, which also discusses the advantages and limitations of the proposed approach, along with suggestions for future work. Finally, Section 5 concludes the paper by summarizing the key findings and highlighting the potential directions for further study.

2. The proposed GNN-enhanced recursive LSTM network

2.1. Graph representation of steel framed structures

The graph-based representation of steel framed structures can be implemented through two distinct topological paradigms: (1) the "Joint-as-Node" (JAN) configuration and (2) the "Element-as-Node" (EAN) configuration, as shown in Fig. 1. In the JAN approach, beam-column joints are abstracted as graph nodes while structural elements (i.e., beams and columns) are represented as connecting edges. Conversely, the EAN formulation inverts this relationship by modeling structural elements as nodes and their connections as edges. This methodological distinction becomes critical when considering the operational mechanics of GNNs, which primarily perform feature learning through iterative node-level information aggregation and updating. The selection between JAN and EAN representations should align with specific engineering objectives. For example, in assessing progressive collapse regions of

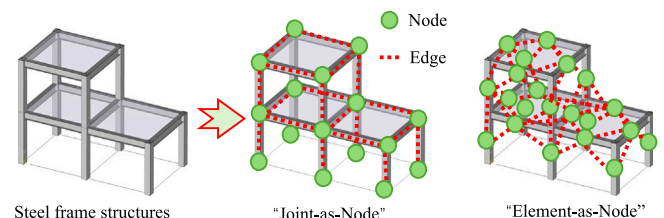


Fig. 1. Graph representation of steel framed structures.

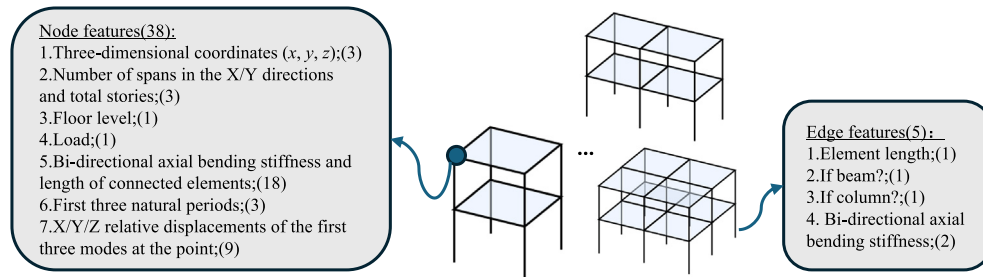


Fig. 2. Schematic illustration of node features and edge features.

structures as conducted by Wang et al. [37], the failure state of elements serves as the prediction target of the GNN, whereas the EAN configuration demonstrates superior performance in this analytical context. While in the structural acceleration and displacement response prediction tasks conducted in this study, given the rigid floor diaphragm assumption typically adopted in structural analysis, where the nodal-level horizontal responses serve as the intended output targets, the JAN configuration would be more appropriate.

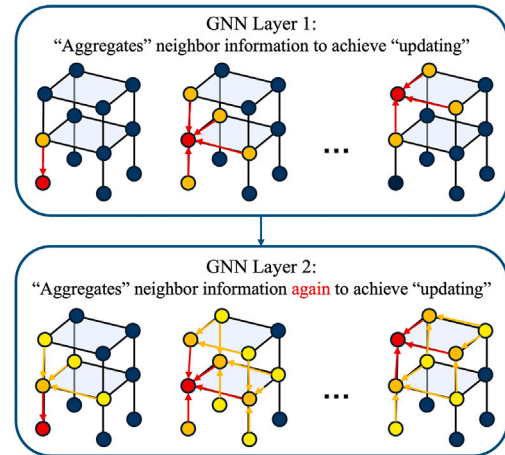
In this study's approach to representing the 3D steel frame as a graph (Fig. 2), each node is assigned a 38-dimensional feature vector encompassing (1) three-dimensional coordinates (x, y, z) (3 dimensions), (2) the number of spans in the X/Y directions and the total number of stories (3 dimensions), (3) the floor level (1 dimension), (4) the load (1 dimension), (5) bi-directional axial bending stiffness and the length of connected elements (18 dimensions), (6) the first three natural periods (3 dimensions), and (7) the X/Y/Z relative displacements of the first three modes at that node (9 dimensions), while each edge is associated with a 5-dimensional feature vector that includes (1) element length (1 dimension), (2) a binary indicator for whether the element is a beam (1 dimension), (3) a binary indicator for whether it is a column (1 dimension), and (4) bi-directional axial bending stiffness (2 dimensions).

Notably, some of the above graph features are partially correlated from a purely mathematical perspective. For example, the two binary indicators “is beam” and “is column” are complementary for the steel frames considered in this study, and the member length or floor level can in principle be inferred from nodal coordinates and graph connectivity. These features were nevertheless retained as physics-informed auxiliary descriptors. The beam/column indicators are adopted as a two-dimensional one-hot representation of the member category, which avoids introducing an artificial ordinal relationship between beams and columns and explicitly distinguishes their different mechanical roles during message passing. Similar graph-based structural engineering studies have also explicitly incorporated member-type or member-length information as structural graph features, although the specific encoding formats may differ. Similarly, although member length and floor index can be derived from nodal coordinates, explicitly providing these features allows the GNN to directly access mechanically meaningful information related to element stiffness and dynamic behavior, rather than requiring the network to infer it from paired nodal coordinates.

For the modal quantities included in the node features, a deterministic sign convention was adopted to address the inherent sign indeterminacy of eigenvectors. Specifically, for each vibration mode, the translational component with the largest absolute amplitude among all nodes was selected as the reference component, and the orientation of the corresponding mode-shape vector was standardized by enforcing this reference component to be positive. The oriented mode-shape vector was then normalized by its maximum absolute translational amplitude.

2.2. Message-passing graph neural network

Message-passing graph neural networks (GNNs) have established themselves as an effective class of neural networks for processing and



* Aggregation of more distant nodes is achieved by deepening the number of GNN layers.

Fig. 3. GNNs enable message passing between nodes through “aggregation” and “update” mechanisms.

learning from graph-structured data. They are designed to address the limitations of conventional neural architectures in modeling relational dependencies by explicitly encoding structural information through node embeddings and edge connections. While deep learning models such as Convolutional Neural Networks (CNNs) excel at processing grid-structured data, including images and sequences, their direct application to non-Euclidean graph domains remains constrained by the irregular connectivity and variable neighborhood structures of graphs. GNNs address these limitations through neighborhood aggregation mechanisms that recursively propagate and transform node features, enabling effective learning of distributed graph representations. This distinctive capability has propelled successful applications across diverse domains, including but not limited to: molecular characterization, social networks, and structural topological representation.

The theoretical foundation of GNNs rests upon an iterative message-passing framework. Formally, this can be expressed as:

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{AGGREGATE} \left(\{ h_u^{(l)} \mid \forall u \in \mathcal{N}(v) \} \right) \right) \quad (1)$$

where $h_v^{(l)} \in \mathbb{R}^d$ denotes the d -dimensional representation of node v at layer l , and $\mathcal{N}(v)$ represents its neighborhood. Stacking L message-passing layers enables each node to integrate information from its L -hop neighborhood (as shown in Fig. 3), establishing a hierarchical abstraction mechanism that facilitates the learning of both local structural patterns and global topological properties. Unlike classical graph algorithms that rely on handcrafted heuristics, GNNs automatically optimize their parameters through gradient-based learning, enabling adaptive discovery of latent relational patterns across various graph topologies and task specifications.

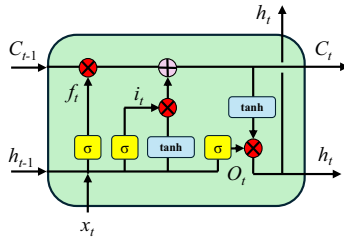


Fig. 4. The illustration of the LSTM cell.

2.3. Recursive LSTM network and corrective training strategy

2.3.1. Recursive LSTM network

Long Short-Term Memory (LSTM) networks are a class of recurrent neural networks (RNNs) that excel at modeling sequential data by capturing long-range dependencies and temporal dynamics. The core innovation of LSTM lies in its gating mechanisms, which regulate the flow of information through a memory cell, enabling the network to selectively retain or discard historical context. The LSTM architecture consists of three primary gates: the input gate, forget gate, and output gate, each of which modulates the flow of information through the network. As illustrated in Fig. 4, the LSTM network operates through the following mathematical formulations:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{Forget gate}) \quad (2)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{Input gate}) \quad (3)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (\text{Candidate memory}) \quad (4)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (\text{Memory cell update}) \quad (5)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (\text{Output gate}) \quad (6)$$

$$h_t = o_t \odot \tanh(C_t) \quad (\text{Hidden state}) \quad (7)$$

where σ and $\tanh$ respectively denote sigmoid and hyperbolic tangent activation functions, $\odot$ indicates element-wise multiplication, and $[\cdot]$ represents vector concatenation. The forget gate f_t dynamically modulates historical memory retention, while the input gate i_t controls new feature integration through the candidate memory $\tilde{C}_t$. The output gate o_t subsequently filters the cell state C_t to produce the hidden state h_t .

In the LSTM model, the memory cell maintains a persistent state that accumulates historical context, enabling the network to model long-term dependencies. Nevertheless, when processing seismic ground motion data characterized by long-sequence attributes encompassing thousands of time steps (often exceeding tens of thousands of tokens), LSTM networks encounter significant challenges. Drawing on the finite difference discretization of time and step-by-step forward calculations in dynamic numerical integration methods for structural seismic response analysis (e.g., the central difference method), wherein the structural response at the subsequent time step is deterministically derived from preceding acceleration, displacement, and external force, an LSTM model can be employed to surrogate this stepwise process. Then, through an iterative extrapolation mechanism, the complete structural response under external seismic excitation sequences can thereby be predicted, forming the recursive LSTM framework adopted in this study. Notably, the term “recursive” in this study refers to the autoregressive prediction strategy, in which the trained LSTM predictor is repeatedly called and previously predicted responses are fed back to construct the complete response history, rather than to the intrinsic recurrent architecture of the LSTM cell itself. In this form, the LSTM model circumvents the challenge of one-time processing of long-sequence data and has better adaptive capability. Accordingly, the construction of training datasets for recursive models generally requires sampling and reconstruction of seismic excitation and structural response time histories at a certain

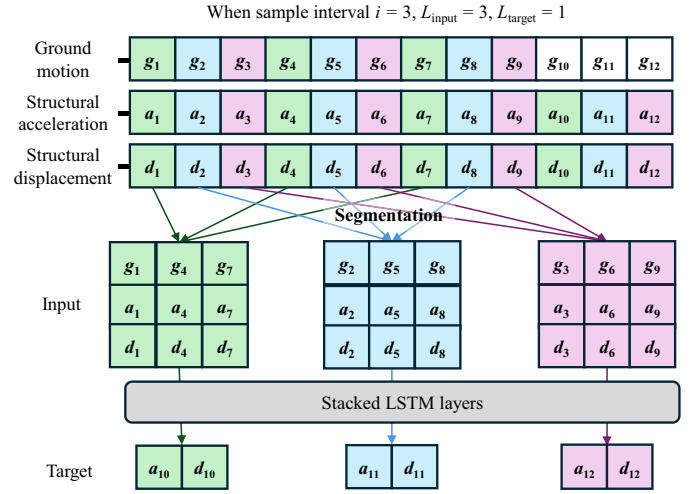


Fig. 5. Example of a typical time-slicing and single-step prediction process.

sampling interval i . Each reconstructed dataset typically comprises a certain length L_{input} of past seismic excitations and structural acceleration and displacement responses, which are used to predict future length L_{target} structural responses. An example of the typical time-slicing and single-step prediction process is presented in Fig. 5.

2.3.2. Corrective training strategy

The above LSTM-based predictor is deployed through the recursive rollout mechanism described in Section 2.3.1, analogous to a step-by-step integration method, to address time series problems—thereby providing an effective solution for seismic response prediction across arbitrary lengths and sampling rates. However, it is noteworthy that this mechanism often tends to diverge in practical applications due to the cumulative errors inherent in the extrapolation process, as observed in related studies [43,44]. This issue primarily arises because the model is trained using ground truth values, whereas during actual use, it relies on predictions that already contain errors to infer subsequent responses. Although optimizing the model by adjusting the input sequence length, sampling interval, and hyperparameter settings is a potential solution [27], such trial-and-error methods can be impractical when the training cost is high. Consequently, this study adopts a corrective training strategy that utilizes a two-stage training process [44] to enhance prediction accuracy and mitigate error accumulation.

Specifically, in the first training stage the model is trained using typical input–output data pairs generated by time-series segmentation, as illustrated in Fig. 5. Once a predetermined number of epochs is reached or the desired accuracy is achieved, the second stage of training is initiated, during which data produced from the model’s own predictions is used for further refinement. For example, when the model is designed to predict the response at the next time step based on L_{input} previous time steps of seismic input and response, the data for the second stage—as depicted in Fig. 6—consists of a sequence longer than $2 \times L_{\text{input}}$. In this sequence, the first L_{input} values remain the ground truth; using these true values along with the first-stage-trained model, recursive prediction is performed step by step. Moreover, gradient updates are applied at each time step rather than after each batch, thereby continuously correcting the model’s prediction bias [44].

2.4. GNN-enhanced recursive LSTM network

The efficacy of such a recursive architecture in predicting response tasks has been demonstrated for a specific single structure [27]. However, in order to extend the applicability of the model to structures

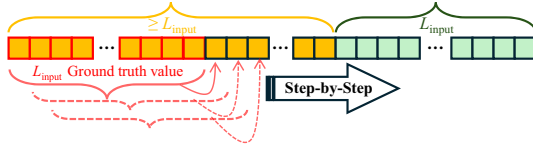


Fig. 6. The illustration of the data for the second training stage.

with different geometric configurations, additional structural information inputs to the network architecture are required. Considering the limitations of expressing 3D framed structures in the form of ordinary tensors, graph representation and GNNs offer viable ideas to fulfill this need. Therefore, a GNN-enhanced recursive LSTM model is proposed for bidirectional seismic response prediction of steel-framed structures in this study.

As shown in Fig. 7, the proposed model aims to predict future structural responses by integrating three main components: a multi-layer GNN module, a stacked LSTM module, and a story feature tensor. In this network architecture, the 3D steel framed structures are first represented as graphs, and a multi-layer GNN then processes this graph to extract the global geometric and dynamic characteristics, culminating in a global pooling step that produces a latent representation encoding the overall structural properties. Concurrently, the ground motion and structural response histories are segmented into time windows, which are fed into a stacked LSTM network so that temporal dependencies can be captured. The final hidden state from the top LSTM layer thus preserves critical time-series features. Meanwhile, the story feature tensor is processed by a multi-layer perceptron (MLP). Afterward, their outputs are concatenated and passed through a final MLP to yield the predicted structural response in future time steps. By combining GNN, LSTM, and the story feature tensor in this way, the model leverages global structural information, temporal dynamics, and local floor-level features simultaneously.

Table 1

Geometric parameters used for layout generation.

Parameter	Level							
	L1	L2	L3	L4	L5	L6	L7	L8
N_{story}	2	3	4	5	6	7	8	9
N_{spanX}	2	3	4	5	6	7	–	–
N_{spanY}	2	3	4	5	6	7	–	–
H_{story} (mm)	2800	2900	3000	3100	–	–	–	–
L_{spanX} (mm)	3000	3600	4200	4800	5400	6000	6600	7200
L_{spanY} (mm)	3000	3600	4200	4800	5400	6000	6600	7200

3. Dataset construction and experiment settings

3.1. Steel framed structures generation

This investigation focuses on regularly configured steel-framed structures, with geometric parameter ranges detailed in Table 1. The number of stories is determined according to the height limitations for pure steel frames (without bracing systems) specified in Chinese steel structure design codes GB 50017–2017 [45]. Span configurations in both orthogonal directions are selected based on practical engineering precedents and the “Guide to Sizing Major Components for Steel-Framed Houses” [46] issued by the Ministry of Housing and Urban-Rural Development of China. All structural geometries in the dataset strictly comply with these codified parameter constraints. For all generated frames, the beam-to-column joints were assumed to be rigid moment connections, and the column bases were assumed to be fixed. This assumption is consistent with the pure steel moment-resisting frame system considered in this study, in which the lateral resistance is primarily provided by beams, columns, and their moment-resisting connections. Since this study focuses on global seismic response prediction rather than connection-level behavior, specific construction details of the joints, such as welded, welded-bolted, or fully bolted details, were not further distinguished. While prior studies predominantly employ

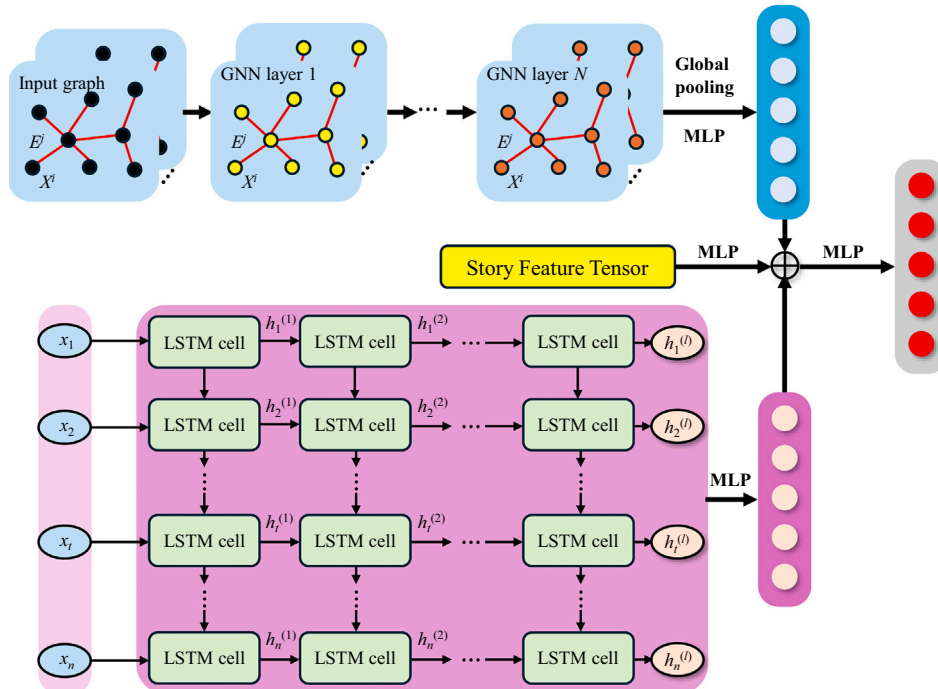


Fig. 7. The architecture of the proposed GNNrLSTM network.

Monte Carlo sampling or Latin hypercube designs for automated 2D steel frame modeling (e.g., OpenSees workflows [47]), the combinatorial complexity of 3D frame design remains unresolved. Critical challenges include: 1) Spatial variability of beam/column member sizes within a single floor; 2) Absence of automated design tools for multi-variable 3D frames.

To address this challenge, an orthogonal experimental design methodology was adopted to obtain balanced and controlled geometric diversity under a limited number of engineering-realistic design cases. Based on the predefined mixed-level geometric factors listed in Table 1, 64 distinct geometric configurations were generated. In this study, each geometric layout required subsequent code-compliant structural design, e.g., section selection. In the absence of a mature automated design tool for multi-variable 3D steel frames, the number of structural schemes that can be generated through manual design is inevitably limited. The selected factors, including the number of stories, the number of spans in the two horizontal directions, story height, and span lengths in the two horizontal directions, directly affect the structural topology, geometric scale, plan aspect ratio, member layout, mass and stiffness distributions, and dynamic characteristics of three-dimensional steel frames. In an orthogonal experimental design, the levels of different factors are arranged according to an orthogonal array, so that the prescribed levels of the main factors are sampled in a relatively balanced manner and the potential coupling among different geometric factors is reduced within a limited number of cases. Compared with arbitrary sampling using the same number of cases, this strategy reduces the risk that the samples are concentrated around only a few specific structural prototypes. In this way, the generated layouts include frames with different heights, span arrangements, plan dimensions, and member lengths within the predefined design ranges. Such a sampling strategy helps ensure that the model is evaluated on a controlled set of structures with different graph topologies and mechanical characteristics, rather than on repeated variants of a single structural prototype. It should be emphasized that this balanced sampling does not imply exhaustive coverage of the full design space of three-dimensional steel frames. Instead, the purpose is to construct an engineering-realistic and computationally manageable database for examining whether the proposed GNNrLSTM can learn shared structural representations across multiple structurally different configurations within the predefined parameter domain.

Notably, during the structural design, the building site is assumed to be located in the Yangpu District, Shanghai, with structural design parameters referenced from Table 2. The standard floors are divided according to design guidelines [46], selecting either two or three floors as the same standard floor based on the total number of stories. Within each standard floor, beams are grouped and designed as exterior beams in the x-direction, exterior beams in the y-direction, interior beams in the x-direction, and interior beams in the y-direction, while columns are categorized and designed as corner columns, exterior columns in the x-direction, exterior columns in the y-direction, and interior columns. All beams employ I-shaped steel sections, and all columns use square steel tube sections. The structural design is conducted using the design software YJK, requiring designers to optimize the design scheme by selecting smaller cross-sections as much as possible while complying with code requirements, including the inter-story drift limit of 1/250 for steel frames under frequent earthquakes. This procedure is consistent with the practical design logic of reducing material usage while satisfying the required structural performance. Ultimately, two structural schemes using Q235 and Q355 steel grades respectively, are developed for each geometric layout, resulting in a total of 128 structural design solutions. Therefore, the variability of the structural database mainly originates from the geometric layouts listed in Table 1, steel grades, standard-floor grouping, and location-dependent member section design. The generated database was constructed within commonly recommended dimensional ranges and through a practical engineering design procedure, making it representative of realistic regular steel moment-resisting frames considered in

Table 2
Structural design data.

Parameter	Value
Seismic intensity level	7
Site category	10
Seismic design group	Group 2
Site characteristic period T_g (s)	0.25
Frame seismic performance level	III
Floor dead load (kN/m ²)	5
Floor live load (kN/m ²)	2

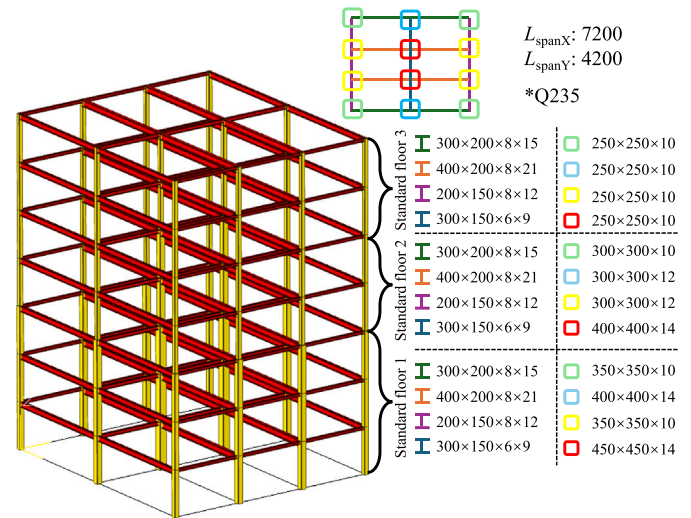


Fig. 8. The illustration of one of the design solutions.

this study, although it is not intended to exhaustively cover all possible steel-frame systems. One of the design cases is presented in Fig. 8.

3.2. Ground motion selection

The seismic resilience evaluation method provided in GB/T 38591–2020 [1] primarily focuses on structural performance under the maximum considered earthquake (MCE). Accordingly, the ground motions are selected from the PEER NGA-West2 database based on their compatibility with the MCE response spectrum. Although this study assumes specific site conditions as previously described, no screening criteria (e.g., magnitude, epicentral distance, or shear wave velocity) are imposed to enhance model generalization capabilities by increasing the diversity and dispersion of ground motions. The sole selection constraint requires ground motion durations exceeding 15 s. This approach yielded 200 pairs of bidirectional ground motion data, with the second component (H2) scaled to 85% of the peak acceleration value of the first component (H1). As shown in Fig. 9, the average response spectrum of selected records is statistically consistent with the MCE target spectrum, that is, their error at the main natural periods of the structures is less than 20%, which meets the requirements of GB/T 38591–2020.

3.3. Nonlinear dynamics time history analysis

The three-dimensional models were developed in the Open System for Earthquake Engineering Simulation (OpenSees) simulation platform [48], where frame members were modeled using nonlinear force-based beam-column elements incorporating *Steel02* material to consider the effect of isotropic strain hardening. As illustrated in Fig. 10, plastic hinge behavior was simulated through such distributed plasticity elements with an assumed plastic hinge length L_p , employing member-specific integration schemes: the “HingeRadau” method was implemented for

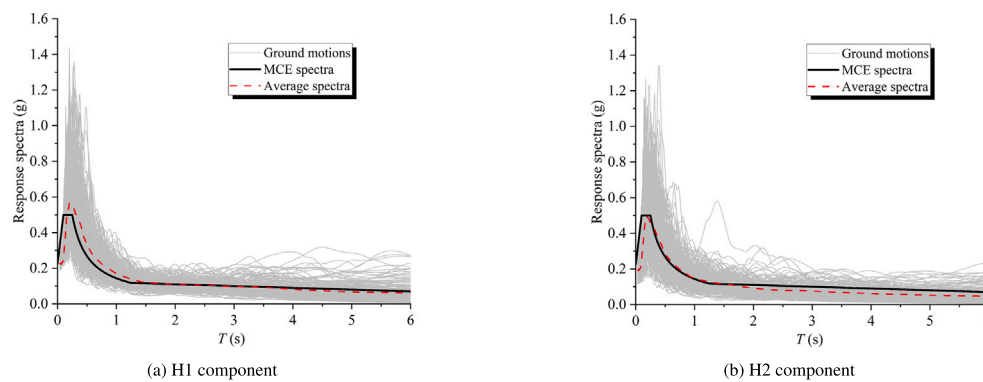


Fig. 9. The response spectrum of selected ground motions.

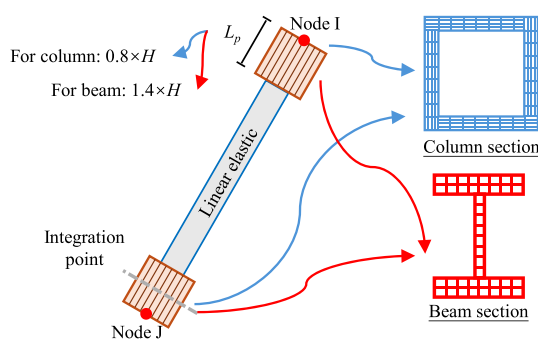


Fig. 10. The illustration of the beam-column element.

steel beams to address concentrated flexural demands near member ends under cyclic loading, while steel columns utilized the “HingeMidpoint” scheme with L_p explicitly defined as 0.8 times the section depth [49]. For the fiber discretization of the I-beam cross-section, 8 fibers are discretized along the longitudinal direction of both the web and flanges, while 1 and 2 fibers are respectively allocated along the thickness direction of the web and flanges. For square steel tubular columns, discretizing 8 fibers along the flange direction and 4 fibers along the thickness direction is reasonable [49]. The Rayleigh damping was adopted and determined by the first and third natural periods. According to GB/T 38591-2020, Appendix B, Item B.2.7 (b), the damping ratio of the steel frame under MCE level can be taken as 0.05. Beams and columns sharing the same joint were modeled using common nodes to implement the rigid moment connection assumption. The horizontal translation degrees of freedom of each node at each floor are constrained based on the rigid diaphragm assumption.

For the nonlinear time-history analyses, the Newmark average acceleration method with $\gamma = 0.5$ and $\beta = 0.25$ was adopted for time integration, and the analysis time step was set to 0.005 s, consistent with the time interval of the ground-motion records. The convergence test was based on the EnergyIncr criterion with a tolerance of 1.0×10^{-6} . To improve numerical robustness for highly nonlinear cases, an adaptive strategy was employed, in which the convergence tolerance could be relaxed to 1.0×10^{-5} , the maximum number of iterations could be increased from 100 to 500, and the time step could be subdivided to a minimum of 0.001 s when convergence difficulties occurred.

The adopted finite element modeling strategy follows the fiber element-based frame modeling paradigm recommended by the Chinese Standard GB/T 38591-2020 [1], making it consistent with the resilience design-oriented objectives of this work. In addition, the steel column modeling strategy is consistent with the experimentally validated approach of Suzuki and Lignos [49], which has been shown to reasonably

reproduce the flexural strength and stiffness deterioration of hollow structural section columns. Therefore, the adopted modeling strategy is supported by both codified guidelines and validated component-level modeling methods, providing a practical and engineering-relevant basis for the nonlinear seismic response analyses conducted in this study.

3.4. Experiment settings

The original dataset was constructed by conducting 4000 non-redundant finite element analyses (FEA), which were generated through random sampling and combination of 128 structural designs and 200 pairs of bidirectional ground motions. The dataset was shuffled using a fixed pseudo-random seed before splitting, and the structural-ground-motion combinations were randomly mixed across the training, validation, and test subsets. It was subsequently partitioned into training (70%, 2800 samples), validation (20%, 800 samples), and test sets (10%, 400 samples). Under this partitioning strategy, the training, validation, and test sets contain different structural-ground-motion combinations sampled from the same predefined pool of structural designs. Therefore, the evaluation primarily reflects the model’s capability to learn shared structural representations among the sampled configurations and to predict their responses under unseen seismic excitations, rather than its extrapolation capability to arbitrary unseen geometric layouts. Following the input-output configuration and training strategy outlined in Section 2.3, the seismic motion and response data for each subset were processed as follows: during preprocessing, a sampling interval of 4 was applied, with the model’s input time-step length set to 100 and the prediction target spanning 1 time-step. In the second training stage described in Section 2.3.2, each data sequence was extended to a length of 300 time-steps, where the first 100 real-time steps were used to drive the model in a step-by-step manner for bias correction. This stage was initiated only after the model achieved an average R^2 of 90% on the validation set across one epoch. This 90% threshold ensures the model captures foundational structural representations while avoiding excessive pursuit of fitting “teacher-forced” (ground-truth) inputs. Although higher Stage 1 performance is attainable through prolonged training, it cannot efficiently address the core issue of error accumulation. In the second stage, the model training employed an early stopping strategy, halting the process when validation performance plateaued over 8 epochs to prevent overfitting. Both the 3D structural graph representations and temporal data were stored using the PyTorch Geometric (PyG) framework’s “Data” class for efficient graph neural network processing.

The GNN-enhanced recursive LSTM network presented in Section 2.4 is utilized to process heterogeneous input data. The proposed GNN-enhanced recursive LSTM network employs four distinct GNN architectures (GAT [50], GIN [51], GCN [52], and GraphSAGE [53]), each configured with 3 layers to ensure consistent depth. All architectures share a core processing pipeline: (1) **Layer-wise processing**: A two-layer MLP with ReLU activation and a hidden dimension of 256 is

applied in each layer to transform node features. (2) **Global aggregation**: Node embeddings are aggregated via global pooling into a latent graph representation (dimension: 128), followed by LayerNorm for stabilization.

Key architectural differences across the four models are as follows:

- **GAT (Graph Attention Network)** incorporates 4 attention heads in each layer to capture multi-faceted node interactions. The outputs of the attention heads are concatenated and projected to the hidden dimension before MLP processing.
- **GIN (Graph Isomorphism Network)** directly applies the MLP-based design described above, emphasizing message-passing with explicit nonlinear transformations between nodes.
- **GCN (Graph Convolutional Network)** simplifies the architecture by replacing MLPs with linear projections and symmetric adjacency matrix convolutions. ReLU and LayerNorm are retained post-convolution to introduce nonlinearity and numerical stability.
- **GraphSAGE (Sample-and-Aggregate)** uses a mean aggregation strategy across neighbors, combined with the same MLP structure (ReLU + 256 hidden units) in each layer to generate node embeddings.

Concurrently, story features (dimension: 38) are transformed through an MLP with ReLU activation and LayerNorm to match the latent dimension (128). The LSTM module consists of 5 layers with a hidden dimension of 128, processing input sequences of dimension 6. The final LSTM output is projected to a 64-dimensional vector via a linear layer with ReLU and LayerNorm. The latent graph representation, processed story features, and LSTM output are concatenated and fed into a final three-layer MLP (128 → 64 → 4) for prediction.

The experiments were conducted on an NVIDIA RTX 4090 24GB GPU and an AMD EPYC 7502 CPU. For training, the model is optimized using the Adam optimizer with an initial learning rate of 1e-3, paired with a “ReduceLROnPlateau” scheduler that reduces the learning rate by a factor of 0.5 after 5 epochs of no improvement, with a minimum learning rate of 1e-4. The model is trained with a batch size of 512, using mean squared error (MSE) loss.

4. Results and discussions

4.1. Performance of the proposed model

The performance of the model was evaluated using time-series R^2 , peak-value R^2 , normalized root mean square error (NRMSE), and relative peak error (RPE), as defined in Eqs. (8)–(11). These metrics were calculated based on the recursive prediction results on the test set rather than the one-step prediction results using true response inputs. The time-series R^2 and NRMSE evaluate the overall time-history prediction accuracy, while the peak-value R^2 and RPE assess the peak response prediction accuracy.

$$\text{Time-series } R^2_{\text{target}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\sum_{t=1}^{T_i} (\hat{y}_{i,\text{target}}^t - y_{i,\text{target}}^t)^2}{\sum_{t=1}^{T_i} (y_{i,\text{target}}^t - \text{mean}(y_{i,\text{target}}))^2} \right) \quad (8)$$

$$\text{Peak-value } R^2_{\text{target}} = 1 - \frac{\sum_{i=1}^N (\hat{y}_{i,\text{target}}^{\text{peak}} - y_{i,\text{target}}^{\text{peak}})^2}{\sum_{i=1}^N (y_{i,\text{target}}^{\text{peak}} - \text{mean}(y_{i,\text{target}}^{\text{peak}}))^2} \quad (9)$$

$$\text{NRMSE}_{\text{target}} = \frac{1}{N} \sum_{i=1}^N \frac{\sqrt{\frac{1}{T_i} \sum_{t=1}^{T_i} (\hat{y}_{i,\text{target}}^t - y_{i,\text{target}}^t)^2}}{\max_t |y_{i,\text{target}}^t|} \times 100\% \quad (10)$$

$$\text{RPE}_{\text{target}} = \frac{1}{N} \sum_{i=1}^N \frac{|\hat{y}_{i,\text{target}}^{\text{peak}} - y_{i,\text{target}}^{\text{peak}}|}{y_{i,\text{target}}^{\text{peak}}} \times 100\% \quad (11)$$

where N is the number of analyses, T_i is the number of time steps in the i -th analysis, $y_{i,\text{target}}^t$ and $\hat{y}_{i,\text{target}}^t$ are the ground-truth and predicted responses, respectively, and $y_{i,\text{target}}^{\text{peak}} = \max_t |y_{i,\text{target}}^t|$ and $\hat{y}_{i,\text{target}}^{\text{peak}} = \max_t |\hat{y}_{i,\text{target}}^t|$.

The model’s performance on the test set is summarized in Table 3. The GraphSAGE model, due to its simpler operational mechanism, exhibited faster training speeds and required fewer epochs. However, its capacity to enhance temporal prediction was limited, likely constrained by its ability to extract high-dimensional structural features. The GIN model performed the best overall, with GAT closely following. Specifically, the GIN model achieved the highest time-series R^2 values for both displacement and acceleration responses, together with the lowest NRMSE values of 4.22% and 3.56%, respectively. In terms of peak response prediction, the GIN model also achieved the highest peak-value R^2 and the lowest RPE values of 3.83% and 4.56% for displacement and acceleration, respectively. These additional error-based metrics indicate that the proposed model not only captures the overall temporal trends of structural responses but also maintains small relative time-history and peak-response errors.

Compared with GIN and GAT, the GCN and GraphSAGE models showed relatively larger NRMSE and RPE values, indicating that their lower R^2 values were accompanied by larger amplitude and peak-response discrepancies. This observation further demonstrates the necessity of supplementing R^2 with error-based metrics when evaluating oscillatory structural responses. Notably, these deep learning models achieved predictions in mere seconds, whereas simulations using OpenSees typically required dozens of minutes, highlighting a significant improvement in computational efficiency. As shown in Fig. 11, trained models accurately captured the bidirectional dynamic response trends of structural systems, though minor discrepancies in amplitude were observed at certain time points compared to ground-truth values.

Fig. 12 illustrates the model performance after completing Stage 1 training. At this point, the model achieved an average R^2 of 0.90 in a single epoch of true-value-based prediction, as seen in the figure: when fed with true-value inputs, the model’s outputs closely align with the actual response at most time steps. However, when directly applied to recursive prediction without further refinement, the model generates significant deviations from the actual conditions, completely diverging from reality. This observation aligns with findings from previous studies

Table 3
Performance (on test set) of different GNN models.

Model	Response	N_{epochs} (stage 1)	Test set performance (stage 2)				Time consumption (h)
			Time-series R^2	Peak value R^2	NRMSE (%)	RPE (%)	
GAT	Disp	55	0.9490	0.9621	4.61	4.23	100.8
	Accel		0.9659	0.9501	4.20	5.09	97.2
GIN	Disp	47	0.9544	0.9683	4.22	3.83	94.5
	Accel		0.9754	0.9562	3.56	4.56	93.4
GCN	Disp	50	0.9132	0.9088	6.88	7.49	94.1
	Accel		0.9144	0.9372	6.47	6.29	97.9
GraphSAGE	Disp	44	0.9034	0.8991	8.21	10.34	92.2
	Accel		0.9154	0.9289	7.42	6.68	92.7

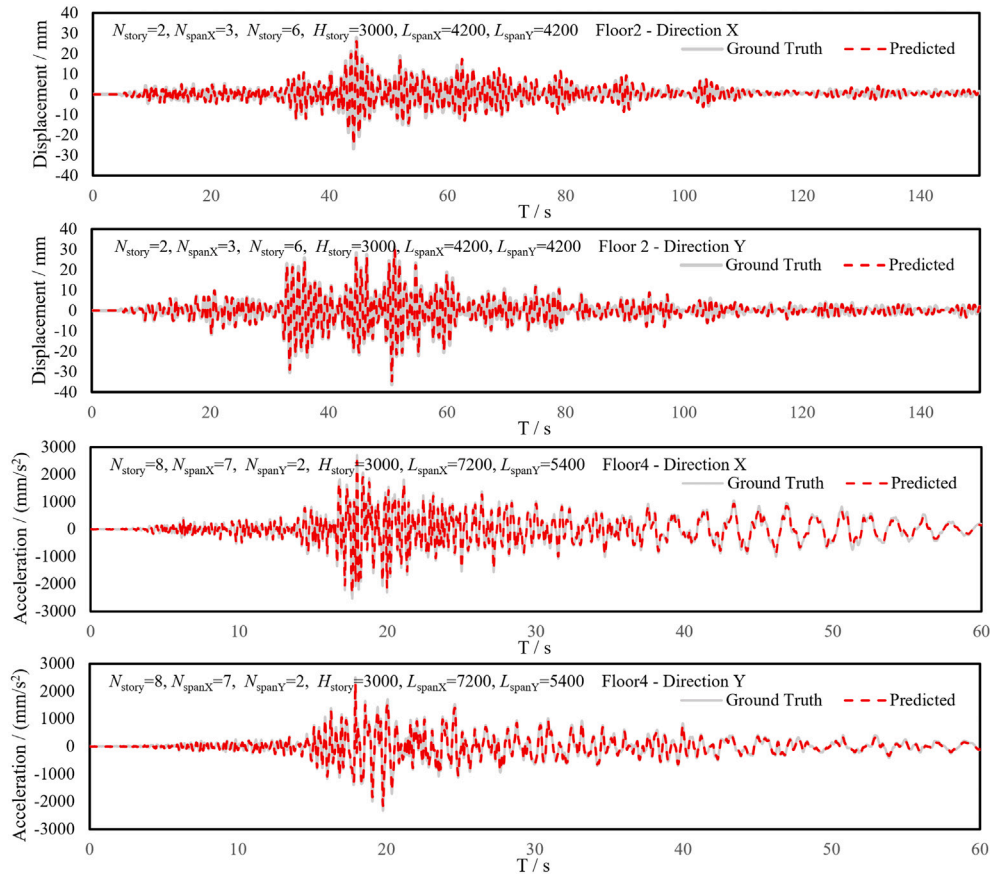


Fig. 11. Some typical comparisons of model predictions to ground truth values.

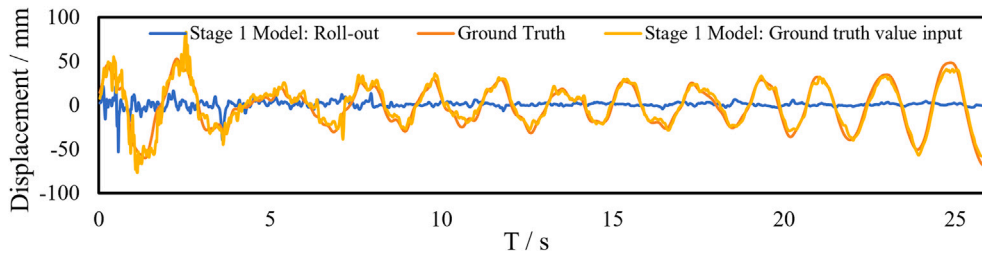


Fig. 12. The performance of the Stage 1 model.

Table 4
Performance of non-GNN model.

Dataset	R^2	Number of epochs
Train set	0.7120	263
Validation set	0.5913	
Test set	0.6022	

[43,44] and underscores the necessity of implementing a second-stage correction strategy to address such discrepancies. The results emphasize that while the initial training stage establishes a foundational accuracy, additional calibration is critical to ensure reliable performance in sequential prediction tasks.

To evaluate the necessity of the GNN component in capturing structural spatiotemporal dynamics, an ablation study was conducted by removing the GNN module from the proposed architecture. The

modified framework retained the story feature inputs and the LSTM-based temporal modeling component but omitted the GNN's graph representation learning capability. To compensate for the absence of structural graph information, global structural parameters (e.g., building span, story height) were explicitly incorporated into the story features. However, despite this adjustment, the model exhibited performance limitations: training R^2 plateaued at approximately 71%, remaining far below the performance achieved by the full GNN-integrated model, as shown in Table 4. Therefore, while explicit inclusion of global structural parameters partially mitigated the loss of GNN-derived features, it failed to capture the nuanced spatial relationships and hierarchical dependencies inherent in 3D structural systems. The GNN's role extended beyond merely increasing model complexity—it served as a critical mechanism for encoding high-dimensional structural topology, enabling the model to effectively integrate spatial and temporal dynamics. This ablation study thus validates the crucial role of the GNN in

enhancing both the representational capacity and predictive reliability of the proposed framework.

4.2. Limitations and future study

This study represents an attempt to leverage GNN for three-dimensional structural seismic response time-series prediction. While demonstrating promising results, the research is not without limitations that warrant further investigation in future work. Specifically, the primary limitations and potential future study directions are outlined as follows:

(1) Insufficient diversity of structures in dataset

Although the 64 geometric layouts were generated using an orthogonal experimental design to provide balanced coverage of the selected geometric factor levels, the current structural database is still limited for fully characterizing the high-dimensional design space of three-dimensional steel frames. This limitation is partly due to the engineering-oriented design procedure adopted in this study: each layout required structural design satisfying both code requirements and practical engineering considerations, while mature automated design tools for multi-variable 3D steel frames are still lacking. Therefore, the present database is more suitable for verifying whether the proposed GNNrLSTM can learn from multiple structurally different configurations and predict their responses under unseen ground motions, rather than for demonstrating robust extrapolation to entirely unseen geometric layouts. When the dataset partitioning was revised to evaluate cross-configuration generalization (i.e., 4000 analyses divided proportionally across 64 geometric configurations, with training sets containing 45 configurations, validation sets containing 13 configurations, and testing sets containing the remaining 6 configurations), a significant performance gap emerged. Even as training R^2 reached 90%, the model's performance on validation and testing sets degraded substantially. This outcome aligns with expectations given the study's design: the three-dimensional structural configurations were generated via orthogonal experimental design, ensuring uniform distribution in the design space while minimizing inter-configuration correlations. Consequently, the graph network struggled to extrapolate effective patterns from limited geometric configurations.

A dataset containing more structural configurations is the key to solving this problem. In this scenario, manual design generation of the dataset by structural engineers becomes somewhat ineffective. Considering the scarcity of publicly available design drawings for reasons such as intellectual property protection, intelligent design methods (e.g., generative design [54,55], reinforcement learning design [56], etc.) may be a promising solution. Through intelligent design, given the geometric configuration of a structure, the generation of design solutions can be done quickly, thus empowering more downstream tasks, such as 3D structural response prediction in this study. Taking the dataset considered in this paper as a reference, the integration of these AI-driven methods could facilitate a transition from manually designed samples with limited geometric layouts to tens of thousands of diverse, code-compliant configurations. This modification would significantly enrich the training set with more diverse—and even irregular—structural layouts, ensuring that the GNN module captures a more exhaustive range of mechanical-topological mappings. Ultimately, this approach would effectively mitigate the model performance reduction and ensure high prediction accuracy across a broader spectrum of previously unseen structural design spaces.

(2) Deeper exploitation of the spatial correlation capabilities of GNN

This study exploits GNNs' high-dimensional representation capability of the overall structural properties, but to a certain extent

ignores GNNs' ability to capture spatial relationships for different nodes, which is GNNs' strong point. Future research should further utilize GNNs' capture of spatial-story-level acceleration and displacement response tasks to the structural component level [41].

(3) Relatively long training hours

Compared to general time-series prediction networks, the addition of GNNs brings more valuable structural properties while also contributing to the increase in the memory occupation of the dataset and the training time of the model, which takes more than 100 hours on the dataset generated from 4000 analyses containing 128 structures. In the future, optimization of the structural graph representation (e.g., simplifying the graphs by considering the rigid diaphragm assumption [40]) and network architecture should be further explored to enhance the potential of the model on larger datasets.

(4) Lack of theoretical analysis

Consistent with many deep learning-based surrogate models in structural dynamics, this study is primarily empirical. A rigorous theoretical analysis on the convergence properties, stability, and error propagation bounds of the proposed GNNrLSTM architecture was not conducted. Future work should aim to establish a theoretical foundation to better understand the model's behavior and guarantee its predictive reliability under various conditions.

4.3. Potential for practical engineering integration

The practical relevance of the proposed model lies in its potential to transform the “time-history analysis” phase within common seismic assessment, performance-based design, and resilience design workflows. By significantly reducing computational time, the model enables engineers to evaluate structural performance under a much larger suite of ground motion records. This overcomes the current limitation where structural analysis often relies on limited record sets due to the prohibitive computational cost of conventional nonlinear time history analysis.

Notably, given that deep learning models are still evolving, their robustness and transparency remain areas for further improvement. Therefore, deep learning models are better positioned as auxiliary design tools for improving efficiency, while final design schemes should still undergo rigorous validation. A relatively appropriate approach that can currently be integrated into resilience-based design workflows is a “surrogate-iteration-validation” framework. In this framework, the novel model serves as an assistive tool for rapid design-space exploration and preliminary optimization, helping to quickly identify design alternatives that satisfy specific resilience objectives. However, the final design solution should preferably be verified through conventional physics-based finite element analysis to ensure compliance with relevant codes. Such a framework enables current design codes to recognize these novel methods as efficiency-enhancing tools during the preliminary design stage, thereby facilitating the wider adoption of large-scale seismic assessment while maintaining strict safety-checking protocols.

5. Summary and conclusions

This paper introduces an effective graph neural network (GNN)-enhanced recursive LSTM architecture for bidirectional seismic response prediction of steel-framed structures. By integrating GNNs to process graphs of three-dimensional structures and generate high-dimensional feature representations, the architecture combines temporal modeling modules and story structural features to predict dynamic behaviors across multiple steel frame configurations. A two-stage training strategy was employed: Stage 1 involved typical training using true-value input-based predictions, while Stage 2 introduced an error-correction mechanism to mitigate iterative prediction drift. Additionally, the performance differences among various GNN modules were analyzed, and

ablation studies validated the necessity of the GNN component. The main findings and conclusions can be drawn as follows:

- The proposed model significantly extends the applicability of recursive LSTM networks by capturing evaluations of acceleration and displacement responses, as well as the peak values in three-dimensional steel frames subjected to bidirectional seismic excitations. Compared to finite element simulations, it achieves engineering-acceptable precision levels while reducing computational time, ensuring its viability for resilience assessment and similar highly computationally demanding tasks.
- Among the four GNN architectures tested on the generated dataset, GIN demonstrated the most notable performance improvement, followed by GAT, while GraphSAGE achieved stable results with the shortest training duration despite lower accuracy.
- The two-stage training framework effectively controlled error accumulation inherent in iterative prediction processes. Directly deploying models trained solely on true-value inputs often led to failure due to error propagation in recursive inference.
- Integrating GNN-derived structural representations outperformed simpler tensor-based structural encoding in enhancing predictive accuracy.

In summary, the proposed framework provides valuable insights and a feasible solution for three-dimensional seismic response prediction of steel structures. It is applicable to computationally intensive applications such as resilience assessment and iterative optimization and could be extended to other structural types (e.g., reinforced concrete frames, braced steel systems) through further research. The study also identifies limitations and future research directions, including enhancing structural diversity in datasets, advancing component-level response prediction, and optimizing graph representation methods. Notably, generating diverse structural configurations via intelligent design methodologies (e.g., generative design, reinforcement learning) is suggested to construct large-scale datasets for validation and optimization of the proposed model, addressing current constraints imposed by limited design efficiency.

CRedit authorship contribution statement

Jiajun Du: Writing—original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wei Wang:** Writing—review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Zhengyang Hou:** Writing—review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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